

Inverse Functions

- 1) Match the graph of the function with the graph of its inverse labeled A, B, C, or D.
- 2) Match the graph of the function with the graph of its inverse labeled A, B, C, or D.
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- 4) Match the graph of the function with the graph of its inverse labeled A, B, C, or D.
- 5) Multiple choice: Determine which function is the inverse of $x + 2y + 3 = 0$.
 - A. $y = -\frac{1}{2}x - \frac{3}{2}$
 - B. $2y + x + 3 = 0$
 - C. $y = -2x - 3$
 - D. $2x - y + 3 = 0$
- 6) Verify that f and g are inverses of each other.
- 7) Find the inverse of $f(x) = x^3 + 4$.
- 8) Find the inverse of $f(x) = \frac{x+1}{5}$.
- 9) Does the function shown in the mapping diagram have an inverse?
- 10) Determine if $f(x) = 3x^{1/3} + 4$ is one-to-one. True or false?
- 11) Determine if $f(x) = \frac{x^2}{x^2 + 1}$ is one-to-one. True or false?
- 12) Multiple choice: Select the function below that is one-to-one.
- 13) Multiple choice: Which function has an inverse?
 - A. $f(x) = |x|$
 - B. $f(x) = x^3, x \leq 0$
 - C. $f(x) = 3x^2 - 1$
 - D. $f(x) = x^2$
- 14) The table shown represents the function f . Use the table to find $f^{-1}(5)$.

x	2	4	5
$f(x)$	3	5	9
- 15) State whether $f(x) = x^2 + 3$ on $(-\infty, \infty)$ has an inverse. If not, explain why.
- 16) State whether $f(x) = 7x$ has an inverse. If not, explain why.
- 17) State whether $f(x) = \sqrt{x} - 4$ has an inverse. If not, explain why.
- 18) State whether $f(x) = |x + 5|$ has an inverse. If not, explain why.
- 19) Given $h(x) = \frac{x+2}{x-5}$, find $h^{-1}(x)$.
- 20) Use the graph of f to find $f^{-1}(6)$.

- 21) Use the graph of f to find $f^{-1}(7)$.
- 22) Use the graph of f to find $f(f^{-1}(7))$.
- 23) Solve $f^{-1}(x) = 3.5$.
- 24) Multiple choice: Given $f(x) = 3x - 2$ and $g(x) = \frac{x}{5}$, find $f(g^{-1}(x))$.
- A. $15x - 4$
 - B. $\frac{3x - 4}{5}$
 - C. $15x - 8$
 - D. $\frac{3x - 2}{5}$
- 25) On the same coordinate plane, sketch the inverse of the function shown.
- 26) Use the graphs of f and g shown to evaluate $f^{-1}(-2)$.
- 27) Evaluate $g^{-1}(0)$.
- 28) Evaluate $g^{-1}(f(-3))$.
- 29) Evaluate $g(f^{-1}(2))$.
- 30) Evaluate $f^{-1}(g^{-1}(0))$.